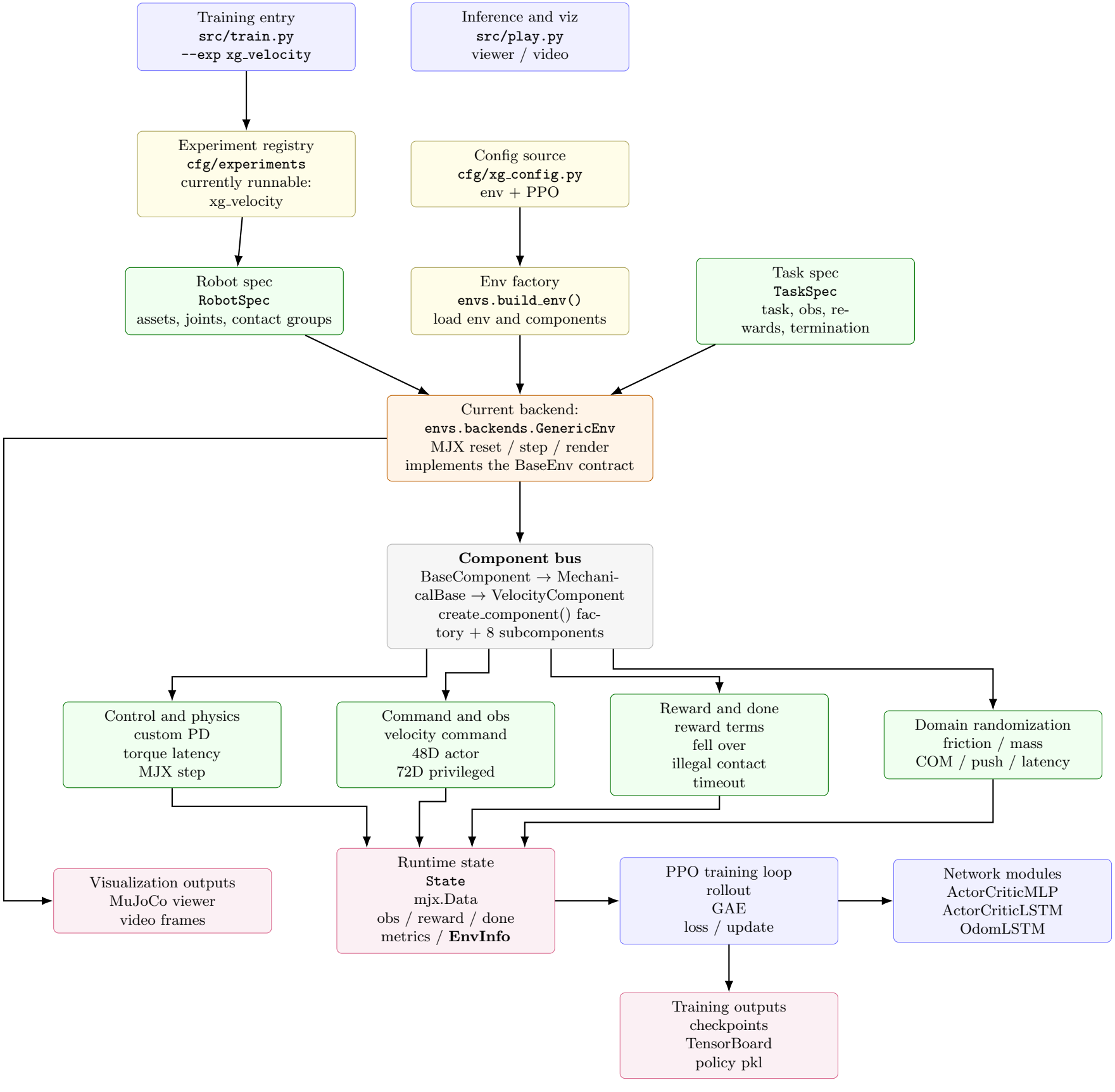
Training entry
src/train.py
--exp xg_velocity
Inference and viz
src/play.py
viewer / video
Experiment registry
cfg/experiments
currently runnable:
xg_velocity
Config source
cfg/xg_config.py
env + PPO
Robot spec
RobotSpec
assets, joints, contact groups
Env factory
envs.build_env()
load env and components
Task spec
TaskSpec
task, obs, re-
wards, termination
Current backend:
envs.backends.GenericEnv
MJX reset / step / render
implements the BaseEnv contract
Component bus
BaseComponent → Mechani-
calBase → VelocityComponent
create_component() fac-
tory + 8 subcomponents
Control and physics
custom PD
torque latency
MJX step
Command and obs
velocity command
48D actor
72D privileged
Reward and done
reward terms
fell over
illegal contact
timeout
Domain randomization
friction / mass
COM / push / latency
Visualization outputs
MuJoCo viewer
video frames
Runtime state
State
mjx.Data
obs / reward / done
metrics / EnvInfo
PPO training loop
rollout
GAE
loss / update
Network modules
ActorCriticMLP
ActorCriticLSTM
OdomLSTM
Training outputs
checkpoints
TensorBoard
policy pkl